

# TRI-NETRA ADCS: design loop, SILS, Monte Carlo and soft OILS

From each customer case to a sized, selected and dispatched attitude-control configuration, verified on the Rust engine (POP v51 in the loop), against the MATLAB twin, and with the flight software running as Cortex-M4F firmware.

|                                                                                                 |                                                                                            |                                                                                                                 |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <div>AIS_3U SELECTED</div> <div><b>mtq_fmr</b></div> <div>closest (not feasible)</div>          | <div>AIS_IMG_3U SELECTED</div> <div><b>mtq_fmr</b></div> <div>closest (not feasible)</div> | <div>ENGINE VS MATLAB</div> <div>VERDICTS</div> <div><b>95/107</b></div> <div>truth environment identical</div> |
| <div>SILS VS SOFT OILS VERDICTS</div> <div><b>105/107</b></div> <div>worst OBC load 6.8 %</div> | <div>VIRTUAL OBC BIT-IDENTICAL</div> <div><b>35</b></div> <div>runs, process + QEMU</div>  | <div>CAMPAIGNS ON THE ENGINE</div> <div><b>8</b></div> <div>MC + edge, vs MATLAB</div>                          |

|                     |                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Owner               | Agastya                                                                                                                               |
| Generated           | 2026-09-28 23:42 UTC                                                                                                                  |
| Repository revision | e81d309                                                                                                                               |
| Engine              | adcs-engine-rs/1.2.0 (adcs-case/1, POP v51 port in-loop, adcs-design, soft OILS)                                                      |
| Flight software     | trinetra-fsw-c/1.0.0 and trinetra-fsw-rs (C99 and Rust no_std, adcs-fswcfg/1)                                                         |
| Cases               | ais_3u, ais_img_3u                                                                                                                    |
| Regenerate          | <code>python3 tools/pipeline.py · python3 tools/engine.py campaign · python3 tools/engine.py oils · python3 tools/vv_report.py</code> |

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## 1. Summary and findings

**ais\_3u** (coarse class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), closest (not feasible), after 4 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 1.600 kg, 0.537 L. Gaps: budget: `mass_kg` 1.6 > 0.35; budget: `volume_L` 0.537 > 0.3

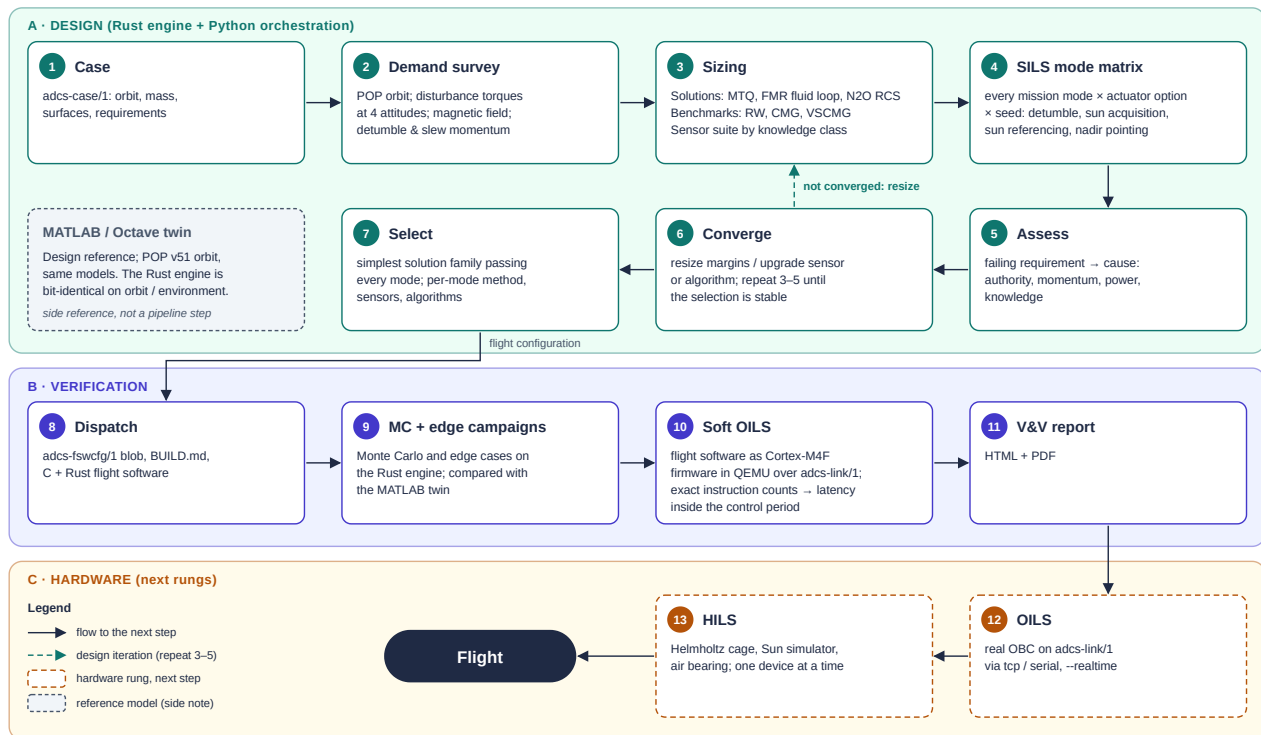
**ais\_img\_3u** (fine class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), closest (not feasible), after 18 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 1.755 kg, 0.534 L. Gaps: budget: `mass_kg` 1.75 > 1

## 2. Architecture and flow

The design loop sizes every actuator option to the case, flies every mission mode with every option in the SILS, converges the sizing, sensors and algorithms, selects the simplest of our solution families that passes, dispatches its flight configuration and verifies it by Monte Carlo and in soft OILS. The same flight software (C99 and Rust, one pseudocode contract) runs in every rung behind one byte-level HAL.

### TRI-NETRA ADCS: from customer case to flight

End-to-end V&V flow: design loop, software verification, then hardware-in-the-loop rungs



TRI-NETRA ADCS - Agastya

Figure 1. From customer case to flight: design loop, verification, hardware rungs.

# Who runs what: languages and layers

TRI-NETRA ADCS, from the flight contract at the top to the ground tooling and outputs at the bottom

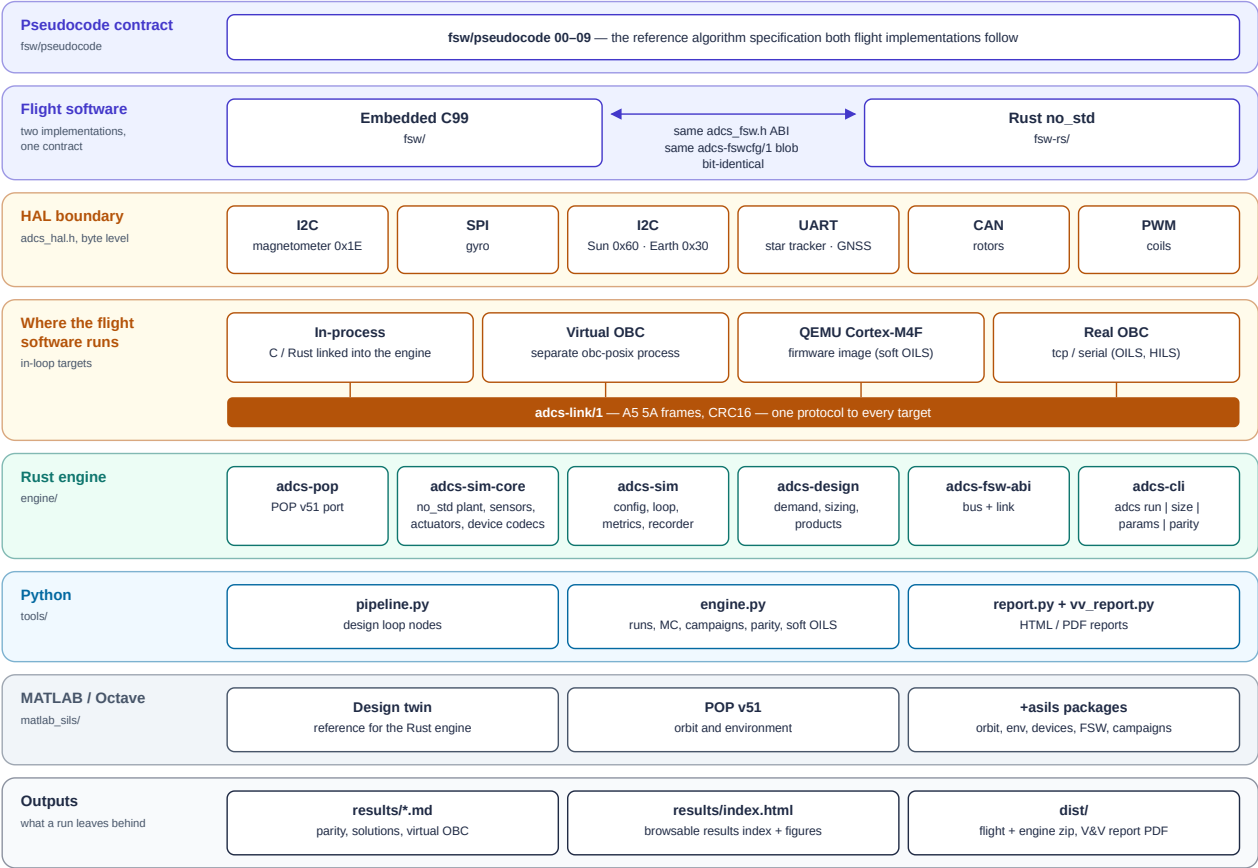


Figure 2. Languages and layers.

### 3. Requirements

#### ais\_3u — 3U AIS satellite — coarse pointing 10 deg, SSO dawn-dusk 550 km

| key            | quantity                             | value  | unit                | source / status                                                                                          |
|----------------|--------------------------------------|--------|---------------------|----------------------------------------------------------------------------------------------------------|
| req.ape        | Absolute pointing error (APE)        | 10.0   | Degree              | stated by the user: the AIS antenna needs only coarse nadir pointing                                     |
| req.ake        | Absolute knowledge error (AKE)       | 5.0    | Degree              | UNCONFIRMED · plan reference: half the APE, so knowledge never dominates the budget                      |
| req.detumble   | Detumble time                        | 284.0  | Minute              | UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec              |
| req.sunacq     | Sun-acquisition time in safe mode    | 95.0   | Minute              | UNCONFIRMED · plan reference: one orbit                                                                  |
| req.mass       | ADCS mass                            | 0.35   | Kilogram            | UNCONFIRMED · plan reference: a coils-and-magnetometer ADCS in a 3U                                      |
| req.pavg       | ADCS orbit-average power             | 0.5    | Watt                | UNCONFIRMED · plan reference: leaves the AIS receiver its share of a 3U power budget                     |
| req.ppk        | ADCS peak power                      | 1.5    | Watt                | UNCONFIRMED · plan reference: all coils at full current                                                  |
| req.vol        | ADCS volume                          | 0.3    | Litre               | UNCONFIRMED · plan reference: under a third of 1U                                                        |
| mission.epoch  | Launch epoch                         | 27.0   | Year                | UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027                                               |
| mission.spd    | Slews per day                        | 10.0   | Count               | UNCONFIRMED · adcs_ref_c1                                                                                |
| mission.sangle | Reference slew angle                 | 30.0   | Degree              | UNCONFIRMED · adcs_ref_c1                                                                                |
| mission.w0     | Initial tumble rate after separation | 10.0   | DegreePerSecond     | UNCONFIRMED · adcs_ref_c1: tumble after separation                                                       |
| orbit.alt      | Altitude                             | 550.0  | Kilometre           | stated by the user (Agastya): 550 km LEO                                                                 |
| orbit.inc      | Inclination                          | 97.593 | Degree              | sun-synchronous at 550 km: asils.orbit.sso_inclination(550)                                              |
| orbit.ecc      | Eccentricity                         | 0.0    | One                 | UNCONFIRMED · adcs_ref_c1                                                                                |
| orbit.ltan     | LTAN                                 | 6.0    | Hour                | stated by the user: SSO dawn-dusk (LTAN 06:00)                                                           |
| mass.m         | Satellite mass                       | 4.0    | Kilogram            | UNCONFIRMED · adcs_ref_c1                                                                                |
| mass.imax      | Largest principal inertia            | 0.042  | KilogramSquareMetre | UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.34^2)$                                        |
| mass.iint      | Intermediate principal inertia       | 0.042  | KilogramSquareMetre | UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal |

| key       | quantity                   | value  | unit                | source / status                                                           |
|-----------|----------------------------|--------|---------------------|---------------------------------------------------------------------------|
| mass.imin | Smallest principal inertia | 0.0067 | KilogramSquareMetre | UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12 \cdot (0.10^2 + 0.10^2)$ |

## ais\_img\_3u — 3U imaging satellite — fine pointing 0.01 deg, SSO 10:00 550 km

| key            | quantity                             | value  | unit            | source / status                                                                                                                              |
|----------------|--------------------------------------|--------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| req.ape        | Absolute pointing error (APE)        | 0.01   | Degree          | stated by the user; 3-sigma. A flown 3U reference: MinXSS-1 (BCT XACT) measured 0.0042-0.0117 deg 3-sigma, axis dependent (arXiv:1706.06967) |
| req.ake        | Absolute knowledge error (AKE)       | 0.005  | Degree          | UNCONFIRMED · plan reference: half the APE                                                                                                   |
| req.rks        | Rate stability                       | 0.001  | DegreePerSecond | UNCONFIRMED · plan reference: 3.6 arcsec/s, to hold smear under an exposure                                                                  |
| req.slew       | Reference slew time                  | 60.0   | Second          | UNCONFIRMED · adcs_ref_c1: mission.sangle in 60 s, target to target                                                                          |
| req.settle     | Settling time after a slew           | 20.0   | Second          | UNCONFIRMED · plan reference: back inside the APE after a slew                                                                               |
| req.detumble   | Detumble time                        | 284.0  | Minute          | UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec                                                  |
| req.sunacq     | Sun-acquisition time in safe mode    | 95.0   | Minute          | UNCONFIRMED · plan reference: one orbit                                                                                                      |
| req.mass       | ADCS mass                            | 1.0    | Kilogram        | UNCONFIRMED · plan reference: wheels, star tracker and coils in a 3U                                                                         |
| req.pavg       | ADCS orbit-average power             | 2.0    | Watt            | UNCONFIRMED · plan reference                                                                                                                 |
| req.ppk        | ADCS peak power                      | 4.0    | Watt            | UNCONFIRMED · plan reference                                                                                                                 |
| req.vol        | ADCS volume                          | 0.6    | Litre           | UNCONFIRMED · plan reference: about half of 1U                                                                                               |
| mission.epoch  | Launch epoch                         | 27.0   | Year            | UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027                                                                                   |
| mission.spd    | Slews per day                        | 10.0   | Count           | UNCONFIRMED · adcs_ref_c1                                                                                                                    |
| mission.sangle | Reference slew angle                 | 30.0   | Degree          | UNCONFIRMED · adcs_ref_c1                                                                                                                    |
| mission.w0     | Initial tumble rate after separation | 10.0   | DegreePerSecond | UNCONFIRMED · adcs_ref_c1: tumble after separation                                                                                           |
| orbit.alt      | Altitude                             | 550.0  | Kilometre       | stated by the user (Agastya): 550 km LEO                                                                                                     |
| orbit.inc      | Inclination                          | 97.593 | Degree          | sun-synchronous at 550 km: asils.orbit.sso_inclination(550)                                                                                  |
| orbit.ecc      | Eccentricity                         | 0.0    | One             | UNCONFIRMED · adcs_ref_c1                                                                                                                    |

| key        | quantity                       | value  | unit                | source / status                                                                                          |
|------------|--------------------------------|--------|---------------------|----------------------------------------------------------------------------------------------------------|
| orbit.ltan | LTAN                           | 10.0   | Hour                | stated by the user: SSO 10:00 (read as LTAN 10:00; LTDN 22:00 gives the same plane and beta magnitude)   |
| mass.m     | Satellite mass                 | 4.0    | Kilogram            | UNCONFIRMED · adcs_ref_c1                                                                                |
| mass.imax  | Largest principal inertia      | 0.042  | KilogramSquareMetre | UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.34^2)$                                        |
| mass.iint  | Intermediate principal inertia | 0.042  | KilogramSquareMetre | UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal |
| mass.imin  | Smallest principal inertia     | 0.0067 | KilogramSquareMetre | UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.10^2)$                                        |

## 4. Design loop per case

### ais\_3u: selected `mtq_fmr` — Magnetorquers + fluid momentum loop (closest (not feasible))

Converged after 4 iteration(s); knowledge class **coarse**; sensors: magnetometer, sun\_sensors, gyro, gnss, earth\_sensor. Demand on the POP orbit: peak disturbance 5.635e-07 N m (sun), stored momentum requirement 0.001466 N m s, torque requirement 5.757e-05 N m, weakest field 19.76  $\mu$ T, detumble momentum 0.00733 N m s.

**Open gaps of the selected family:** budget: mass\_kg 1.6 > 0.35; budget: volume\_L 0.537 > 0.3

| iter | knobs                                                         | options feasible | selection                        | changes for the next iteration                                                                                                                                                                                                                                                                                                                                                                     | blocked |
|------|---------------------------------------------------------------|------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1    | laws as written                                               | 1/25             | mtq (closest (not feasible))     | nadir_pointing/fmr+mtq: power -> electromagnetic pump with more copper (lambda 0.1 -> 0.3 kg/W)<br>nadir_pointing/mtq: fly every mtq_pointing algorithm<br>sun_acquisition/mtq: fly every sun_acquisition algorithm<br>sun_referencing/mtq: fly every mtq_pointing algorithm<br>sun_referencing/vscmg+mtq: fly every pointing algorithm<br>sun_referencing/vscmg+rsc: fly every pointing algorithm | 14      |
| 2    | pump $\lambda$ 0.3 kg/W                                       | 4/25             | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): fluid-loop momentum x1 -> x0.75<br>mtqp: authority x1 -> x1.5 (performance)                                                                                                                                                                                                                                                                                                        | 18      |
| 3    | fmr $\times$ 0.75, mtqp $\times$ 1.5, pump $\lambda$ 0.3 kg/W | 3/25             | mtq (closest (not feasible))     | mtqp: authority back to x1 (no improvement)<br>sun_referencing/fmr+mtq: power -> electromagnetic pump with more copper (lambda 0.3 -> 0.9 kg/W)<br>mass lever scale.fmr undone: it broke<br>sun_referencing/fmr+mtq, requirement violation 0 -> 0.0247                                                                                                                                             | 18      |
| 4    | fmr $\times$ 1, mtqp $\times$ 1, pump $\lambda$ 0.9 kg/W      | 6/25             | mtq_fmr (closest (not feasible)) | —                                                                                                                                                                                                                                                                                                                                                                                                  | 18      |

Why the loop stopped (nothing left that a knob can change):

- detumble/rsc: power fails at the sized authority (rsc); the part's standby power is the floor
- mass (mtq\_fmr): no lever left (budget: mass\_kg 1.6 > 0.35 budget: volume\_L 0.537 > 0.3)
- mtqp: more authority did not reduce the performance violation (39.9 -> 41); kept at x1
- nadir\_pointing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir\_pointing/cmg+rsc: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir\_pointing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor
- nadir\_pointing/rw+rsc: power fails at the sized authority (rw); the part's standby power is the floor
- nadir\_pointing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- nadir\_pointing/vscmg+rsc: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun\_acquisition/cmg: power fails at the sized authority (cmg); the part's standby power is the floor
- sun\_acquisition/rw: power fails at the sized authority (rw); the part's standby power is the floor
- sun\_acquisition/vscmg: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun\_referencing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- sun\_referencing/cmg+rsc: power fails at the sized authority (cmg); the part's standby power is the floor
- sun\_referencing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor
- sun\_referencing/rw+rsc: power fails at the sized authority (rw); the part's standby power is the floor
- sun\_referencing/vscmg+mtq: performance and power both fail — no authority change helps

- sun\_referencing/vscmg+rscs: power fails at the sized authority (vscmg); the part's standby power is the floor

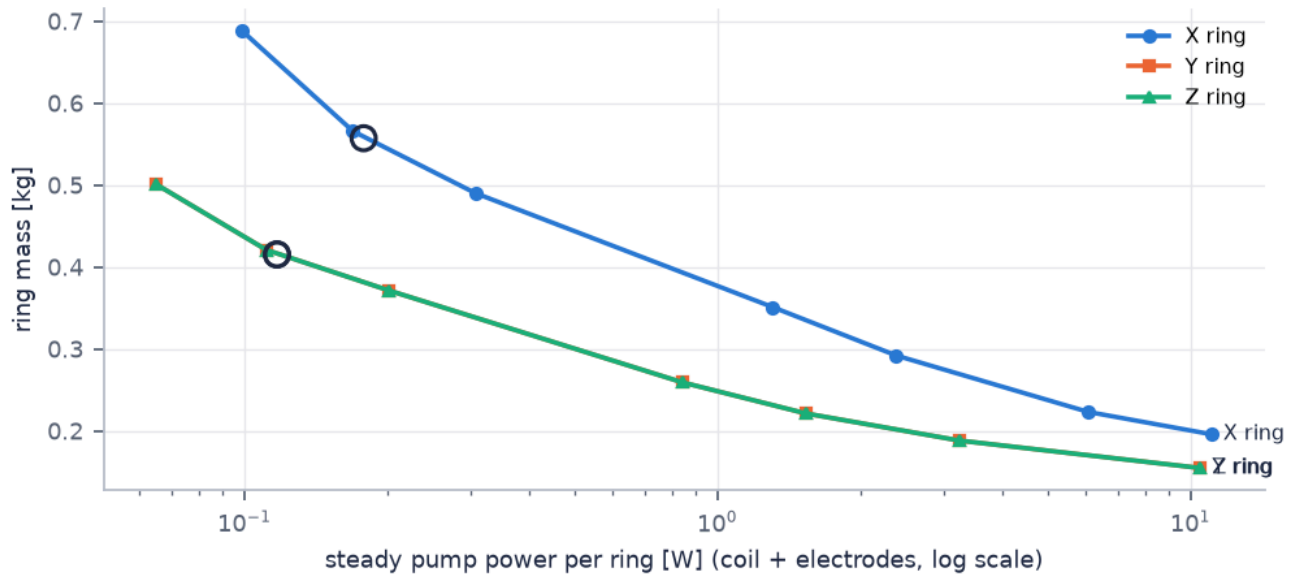
Every family, scored the same way (benchmarks for comparison only):

| family        | role      | feasible | mass [kg] | power [W] | volume [L] | gaps                                                                                                                                                                        |
|---------------|-----------|----------|-----------|-----------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mtq           | solution  | FAIL     | 0.236     | 2.06      | 0.0183     | sun_acquisition: sun_acquisition_time (performance), sun_angle_p95 (performance); sun_referencing: sun_ape_p9973 (performance); nadir_pointing: ape_los_p9973 (performance) |
| mtq_fmr       | solution  | FAIL     | 1.6       | 2.24      | 0.537      | budget: mass_kg 1.6 > 0.35; budget: volume_L 0.537 > 0.3                                                                                                                    |
| mtq_fmr_rcs   | solution  | FAIL     | 2.12      | 2.29      | 1.52       | budget: mass_kg 2.12 > 0.35; budget: volume_L 1.52 > 0.3                                                                                                                    |
| mtq_rw        | benchmark | FAIL     | 0.476     | 2.56      | 0.141      | sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 0.476 > 0.35                                  |
| mtq_rw_rcs    | benchmark | FAIL     | 0.995     | 2.61      | 1.12       | sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 0.995 > 0.35; budget: volume_L 1.12 > 0.3     |
| mtq_cmg       | benchmark | FAIL     | 0.56      | 2.75      | 0.275      | sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 0.56 > 0.35                                   |
| mtq_cmg_rcs   | benchmark | FAIL     | 1.08      | 2.8       | 1.26       | sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 1.08 > 0.35; budget: volume_L 1.26 > 0.3      |
| mtq_vscmg     | benchmark | FAIL     | 0.595     | 3.15      | 0.275      | sun_acquisition: power_mean (power); sun_referencing: sun_ape_p9973 (performance); nadir_pointing: power_mean (power); budget: mass_kg 0.595 > 0.35                         |
| mtq_vscmg_rcs | benchmark | FAIL     | 1.11      | 3.2       | 1.26       | sun_acquisition: power_mean (power); sun_referencing: power_mean (power); nadir_pointing: power_mean (power); budget: mass_kg 1.11 > 0.35; budget: volume_L 1.26 > 0.3      |

Selected method per mission mode:

| mode            | option | feasible | objective (worst seed)     | algorithms                                                                                                                 |
|-----------------|--------|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| detumble        | mtq    | pass     | 48.74 detumble_time        | attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2                                       |
| sun_acquisition | fmr    | pass     | 2.292 sun_acquisition_time | allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2 |

| mode            | option  | feasible | objective (worst seed)  | algorithms                                                                                                                 |
|-----------------|---------|----------|-------------------------|----------------------------------------------------------------------------------------------------------------------------|
| sun_referencing | fmr+mtq | pass     | 0.2513<br>sun_ape_p9973 | allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2 |
| nadir_pointing  | fmr+mtq | pass     | 0.1514 ape_loos_p9973   | allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2 |



Electromagnetic pump design, ais\_3u: each ring's mass against steady power as the mass/power rate  $\lambda$  runs from 0.003 to 3 kg/W; circles mark the converged designs ( $\lambda = 0.9$  kg/W).

Monte Carlo of the dispatched mission (12 runs; inertia, residual\_dipole, cm\_offset, solar\_flux, kp, accommodation, reflectivity, arg\_lat\_deg):

| metric         | req | mean ± std      | range            | pass rate |
|----------------|-----|-----------------|------------------|-----------|
| detumble_time  | 284 | 58.22 ± 22.1    | [9.858, 94.17]   | 100 %     |
| ape_loos_p9973 | 10  | 0.2426 ± 0.0731 | [0.1603, 0.3825] | 100 %     |
| ake_loos_p9973 | 5   | 0.2393 ± 0.0732 | [0.1574, 0.3818] | 100 %     |
| power_mean     | 0.5 | 0.2085 ± 0.0528 | [0.1472, 0.3085] | 100 %     |
| power_peak     | —   | 0.5628 ± 0.0117 | [0.5459, 0.5875] | —         |
| propellant     | —   | 0 ± 0           | [0, 0]           | —         |

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

| metric         | req | SILS        | soft OILS, C | soft OILS, Rust |
|----------------|-----|-------------|--------------|-----------------|
| detumble_time  | 284 | 55.64 pass  | 55.71 pass   | 55.71 pass      |
| ape_loos_p9973 | 10  | 0.1516 pass | 0.1217 pass  | 0.1274 pass     |
| ake_loos_p9973 | 5   | 0.15 pass   | 0.1184 pass  | 0.1234 pass     |
| power_mean     | 0.5 | 0.1593 pass | 0.1501 pass  | 0.1509 pass     |
| power_peak     | —   | 0.5414 —    | 0.5528 —     | 0.5536 —        |
| propellant     | —   | 0 —         | 0 —          | 0 —             |

## ais\_img\_3u: selected `mtq_fmr` — Magnetorquers + fluid momentum loop (closest (not feasible))

Converged after 18 iteration(s); knowledge class **fine**; sensors: star\_tracker, magnetometer, sun\_sensors, gyro, gnss, earth\_sensor. Demand on the POP orbit: peak disturbance 4.367e-07 N m (sun), stored momentum requirement 0.001552 N m s, torque requirement 5.757e-05 N m, weakest field 23.63  $\mu$ T, detumble momentum 0.00733 N m s.

**Open gaps of the selected family:** budget: mass\_kg 1.75 > 1

| iter | knobs                                                                        | options feasible | selection                           | changes for the next iteration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | blocked |
|------|------------------------------------------------------------------------------|------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1    | laws as written                                                              | 13/25            | mtq_fmr<br>(closest (not feasible)) | nadir_pointing/cmg+mtq: fly every pointing algorithm<br>nadir_pointing/cmg+rcs: fly every pointing algorithm<br>nadir_pointing/fmr+mtq: fly every pointing algorithm<br>nadir_pointing/fmr+rcs: fly every pointing algorithm<br>nadir_pointing/mtq: fly every mtq_pointing algorithm<br>nadir_pointing/rw+mtq: fly every pointing algorithm<br>nadir_pointing/rw+rcs: fly every pointing algorithm<br>nadir_pointing/vscmg+mtq: fly every pointing algorithm<br>nadir_pointing/vscmg+rcs: fly every pointing algorithm<br>sun_acquisition/mtq: fly every sun_acquisition algorithm<br>sun_referencing/mtq: fly every mtq_pointing algorithm | 3       |
| 2    | laws as written                                                              | 13/25            | mtq_fmr<br>(closest (not feasible)) | rate stability: gyro noise x1 -> x0.3 (fibre-optic class)<br>mtqp: authority x1 -> x1.5 (performance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4       |
| 3    | mtqp $\times$ 1.5, gyro noise x0.3                                           | 18/25            | mtq_fmr<br>(closest (not feasible)) | mtqp: authority back to x1 (no improvement)<br>rate stability: gyro noise x0.3 -> x0.1 (fibre-optic class)<br>vscmg: authority x1 -> x1.5 (performance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5       |
| 4    | mtqp $\times$ 1, vscmg $\times$ 1.5, gyro noise x0.1                         | 19/25            | mtq_fmr<br>(closest (not feasible)) | nadir_pointing/fmr+mtq: fluid-loop flow sensor 2 -> 0.5 mm/s (1 sigma)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5       |
| 5    | mtqp $\times$ 1, vscmg $\times$ 1.5, flow sensor 0.5 mm/s, gyro noise x0.1   | 20/25            | mtq_fmr<br>(closest (not feasible)) | nadir_pointing/fmr+mtq: fluid-loop flow sensor 0.5 -> 0.125 mm/s (1 sigma)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2       |
| 6    | mtqp $\times$ 1, vscmg $\times$ 1.5, flow sensor 0.125 mm/s, gyro noise x0.1 | 20/25            | mtq_fmr<br>(closest (not feasible)) | nadir_pointing/fmr+mtq: fluid-loop flow sensor 0.125 -> 0.05 mm/s (1 sigma)<br>fmr: authority x1 -> x1.5 (performance)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2       |

| iter | knobs                                                                                                                                                                           | options feasible | selection                        | changes for the next iteration                                                                                                                                                                                                                                                                | blocked |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 7    | mtqp $\times 1$ , vscmg $\times 1.5$ , fmr $\times 1.5$ , flow sensor 0.05 mm/s, gyro noise $\times 0.1$                                                                        | 20/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): gyro noise $\times 0.1 \rightarrow \times 0.3$ (a lighter gyro)<br>fmr: authority $\times 1.5 \rightarrow \times 2.25$ (performance)                                                                                                                                          | 2       |
| 8    | mtqp $\times 1$ , vscmg $\times 1.5$ , fmr $\times 2.25$ , flow sensor 0.05 mm/s, gyro noise $\times 0.3$                                                                       | 19/25            | mtq_fmr (closest (not feasible)) | fmr: authority back to $\times 1.5$ (no improvement)<br>rate stability: gyro noise $\times 0.3 \rightarrow \times 0.1$ (fibre-optic class)<br>mass lever gyro_grade undone: it broke requirement violation 0.538 $\rightarrow$ 1.99                                                           | 4       |
| 9    | mtqp $\times 1$ , vscmg $\times 1.5$ , fmr $\times 1.5$ , flow sensor 0.05 mm/s, gyro noise $\times 0.1$                                                                        | 20/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): one star-tracker head instead of two                                                                                                                                                                                                                                          | 3       |
| 10   | mtqp $\times 1$ , vscmg $\times 1.5$ , fmr $\times 1.5$ , 1 ST head, flow sensor 0.05 mm/s, gyro noise $\times 0.1$                                                             | 13/25            | mtq_fmr (closest (not feasible)) | mass lever st_heads undone: it broke requirement violation 0.538 $\rightarrow$ 2.82<br>cmg: authority $\times 1 \rightarrow \times 1.5$ (performance)<br>rw: authority $\times 1 \rightarrow \times 1.5$ (performance)<br>vscmg: authority $\times 1.5 \rightarrow \times 2.25$ (performance) | 12      |
| 11   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 1.5$ , cmg $\times 1.5$ , rw $\times 1.5$ , flow sensor 0.05 mm/s, gyro noise $\times 0.1$                                  | 20/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): lighter pump, less copper ( $\lambda$ 0.1 $\rightarrow$ 0.0333333 kg/W)                                                                                                                                                                                                       | 3       |
| 12   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 1.5$ , cmg $\times 1.5$ , rw $\times 1.5$ , pump $\lambda$ 0.0333333 kg/W, flow sensor 0.05 mm/s, gyro noise $\times 0.1$   | 19/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): lighter pump, less copper ( $\lambda$ 0.0333333 $\rightarrow$ 0.0111111 kg/W)                                                                                                                                                                                                 | 4       |
| 13   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 1.5$ , cmg $\times 1.5$ , rw $\times 1.5$ , pump $\lambda$ 0.0111111 kg/W, flow sensor 0.05 mm/s, gyro noise $\times 0.1$   | 20/25            | mtq_fmr (closest (not feasible)) | mass lever fmr_lambda undone: it broke sun_acquisition/fmr                                                                                                                                                                                                                                    | 4       |
| 14   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 1.5$ , cmg $\times 1.5$ , rw $\times 1.5$ , pump $\lambda$ 0.0333333 kg/W, flow sensor 0.05 mm/s, gyro noise $\times 0.1$   | 19/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): fluid-loop momentum $\times 1.5 \rightarrow \times 1.125$                                                                                                                                                                                                                     | 4       |
| 15   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 1.12$ , cmg $\times 1.5$ , rw $\times 1.5$ , pump $\lambda$ 0.0333333 kg/W, flow sensor 0.05 mm/s, gyro noise $\times 0.1$  | 19/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): fluid-loop momentum $\times 1.125 \rightarrow \times 0.84375$                                                                                                                                                                                                                 | 4       |
| 16   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 0.844$ , cmg $\times 1.5$ , rw $\times 1.5$ , pump $\lambda$ 0.0333333 kg/W, flow sensor 0.05 mm/s, gyro noise $\times 0.1$ | 19/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): fluid-loop momentum $\times 0.84375 \rightarrow \times 0.632812$                                                                                                                                                                                                              | 4       |
| 17   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 0.633$ , cmg $\times 1.5$ , rw $\times 1.5$ , pump $\lambda$ 0.0333333 kg/W, flow sensor 0.05 mm/s, gyro noise $\times 0.1$ | 21/25            | mtq_fmr (closest (not feasible)) | mass (mtq_fmr): fluid-loop momentum $\times 0.632812 \rightarrow \times 0.5$                                                                                                                                                                                                                  | 3       |
| 18   | mtqp $\times 1$ , vscmg $\times 2.25$ , fmr $\times 0.5$ , cmg $\times 1.5$ , rw $\times 1.5$ , pump $\lambda$ 0.0333333 kg/W, flow sensor 0.05 mm/s, gyro noise $\times 0.1$   | 21/25            | mtq_fmr (closest (not feasible)) | —                                                                                                                                                                                                                                                                                             | 4       |

Why the loop stopped (nothing left that a knob can change):

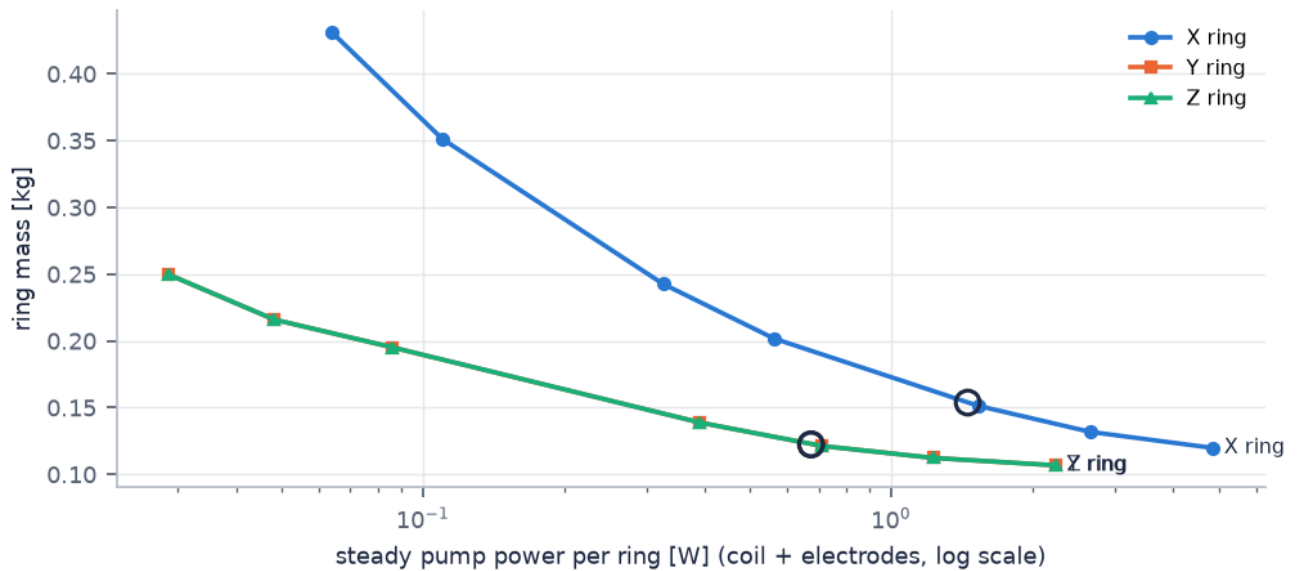
- mass (mtq\_fmr): no lever left (budget: mass\_kg 1.75 > 1)
- mtqp: more authority did not reduce the performance violation (1.35e+04 -> 1.36e+04); kept at x1
- nadir\_pointing/mtq: knowledge fails with the star tracker fitted
- sun\_referencing/fmr+rcs: power is the thrusters' valve power (RCS dumping), not the pump

Every family, scored the same way (benchmarks for comparison only):

| family        | role      | feasible | mass [kg] | power [W] | volume [L] | gaps                                                                                                                                                                                                                                                                 |
|---------------|-----------|----------|-----------|-----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mtq           | solution  | FAIL     | 1.37      | 6.6       | 0.0118     | sun_acquisition: sun_acquisition_time (performance), sun_angle_p95 (performance); sun_referencing: sun_ape_p9973 (performance); nadir_pointing: ape_los_p9973 (performance), ake_los_p9973 (knowledge), rate_stability_p9973 (performance); budget: mass_kg 1.37 > 1 |
| mtq_fmr       | solution  | FAIL     | 1.75      | 9.26      | 0.534      | budget: mass_kg 1.75 > 1                                                                                                                                                                                                                                             |
| mtq_fmr_rcs   | solution  | FAIL     | 2.22      | 9.31      | 1.38       | budget: mass_kg 2.22 > 1; budget: volume_L 1.38 > 0.6                                                                                                                                                                                                                |
| mtq_rw        | benchmark | FAIL     | 1.65      | 7.27      | 0.164      | budget: mass_kg 1.65 > 1                                                                                                                                                                                                                                             |
| mtq_rw_rcs    | benchmark | FAIL     | 2.11      | 7.32      | 1.01       | budget: mass_kg 2.11 > 1; budget: volume_L 1.01 > 0.6                                                                                                                                                                                                                |
| mtq_cmg       | benchmark | FAIL     | 1.73      | 7.44      | 0.316      | budget: mass_kg 1.73 > 1                                                                                                                                                                                                                                             |
| mtq_cmg_rcs   | benchmark | FAIL     | 2.2       | 7.49      | 1.16       | budget: mass_kg 2.2 > 1; budget: volume_L 1.16 > 0.6                                                                                                                                                                                                                 |
| mtq_vscmg     | benchmark | FAIL     | 1.81      | 7.93      | 0.369      | budget: mass_kg 1.81 > 1                                                                                                                                                                                                                                             |
| mtq_vscmg_rcs | benchmark | FAIL     | 2.28      | 7.98      | 1.21       | budget: mass_kg 2.28 > 1; budget: volume_L 1.21 > 0.6                                                                                                                                                                                                                |

Selected method per mission mode:

| mode            | option  | feasible | objective (worst seed)     | algorithms                                                                                                                 |
|-----------------|---------|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| detumble        | mtq     | pass     | 63.54 detumble_time        | attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2                                       |
| sun_acquisition | fmr     | pass     | 2.275 sun_acquisition_time | allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2 |
| sun_referencing | fmr+mtq | pass     | 0.00474 sun_ape_p9973      | allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2 |
| nadir_pointing  | fmr+mtq | pass     | 0.0013 ape_los_p9973       | allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=smc, sun_acquisition=sunspin_l1l2 |



Electromagnetic pump design, ais\_img\_3u: each ring's mass against steady power as the mass/power rate  $\lambda$  runs from 0.003 to 3 kg/W; circles mark the converged designs ( $\lambda = 0.0333333$  kg/W).

Monte Carlo of the dispatched mission (12 runs; inertia, residual\_dipole, cm\_offset, solar\_flux, accommodation, reflectivity, arg\_lat\_deg):

| metric        | req   | mean ± std          | range                 | pass rate |
|---------------|-------|---------------------|-----------------------|-----------|
| detumble_time | 284   | 73.22 ± 15.6        | [44.46, 92.33]        | 100 %     |
| ape_los_p9973 | 0.01  | 0.007674 ± 0.00543  | [0.001366, 0.01746]   | 83 %      |
| ake_los_p9973 | 0.005 | 0.001205 ± 0.000383 | [0.0007565, 0.001958] | 100 %     |
| power_mean    | 2     | 1.173 ± 0.41        | [0.2061, 1.669]       | 100 %     |
| power_peak    | —     | 2.904 ± 0.0111      | [2.886, 2.918]        | —         |
| propellant    | —     | 0 ± 0               | [0, 0]                | —         |

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

| metric        | req   | SILS                 | soft OILS, C         | soft OILS, Rust      |
|---------------|-------|----------------------|----------------------|----------------------|
| detumble_time | 284   | 81.84 <b>pass</b>    | 82.78 <b>pass</b>    | 82.72 <b>pass</b>    |
| ape_los_p9973 | 0.01  | 0.001184 <b>pass</b> | 0.001136 <b>pass</b> | 0.001113 <b>pass</b> |
| ake_los_p9973 | 0.005 | 0.001159 <b>pass</b> | 0.001152 <b>pass</b> | 0.001158 <b>pass</b> |
| power_mean    | 2     | 1.349 <b>pass</b>    | 1.388 <b>pass</b>    | 1.456 <b>pass</b>    |
| power_peak    | —     | 2.897 —              | 2.906 —              | 2.905 —              |
| propellant    | —     | 0 —                  | 0 —                  | 0 —                  |

## 5. SILS scenario verification

Every scenario on the Rust engine with the C flight software behind the byte HAL (POP in the loop). 21 of 40 scenarios meet every requirement they carry; the failures are findings of the design (section 10), not engine errors: the same verdicts come out of the MATLAB twin (section 7).

| scenario              | what                                                        | product          | duration [s] | requirement metrics: value / req                                                                                                                                                |
|-----------------------|-------------------------------------------------------------|------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agile_slew_cmgy       | 3U imaging — (cmgy): 90 deg pitch in 15 s                   | TRN-P-3U-CMG     | 600          | ape_los_on_target_p9973 0.004544 / 0.01 <b>pass</b>                                                                                                                             |
| agile_slew_fmr_rcs    | 3U imaging — (fmr_rcs): 90 deg pitch in 15 s                | TRN-P-3U-FMR-RCS | 600          | ape_los_on_target_p9973 0.006614 / 0.01 <b>pass</b>                                                                                                                             |
| agile_slew_img        | 3U imaging — (img): 90 deg pitch in 15 s                    | TRN-P-3U-IMG     | 600          | ape_los_on_target_p9973 0.008686 / 0.01 <b>pass</b>                                                                                                                             |
| agile_slew_rw_rcs     | 3U imaging — (rw_rcs): 90 deg pitch in 15 s                 | TRN-P-3U-RW-RCS  | 600          | ape_los_on_target_p9973 0.008702 / 0.01 <b>pass</b>                                                                                                                             |
| agile_slew_vscmg      | 3U imaging — (vscmg): 90 deg pitch in 15 s                  | TRN-P-3U-VSCMG   | 600          | ape_los_on_target_p9973 0.01008 / 0.01 <b>FAIL</b>                                                                                                                              |
| detumble_ais          | 3U AIS — B-dot detumble from a 10 deg/s tumble (coils only) | TRN-P-3U-AIS     | 1.722e+04    | detumble_time 37.79 / 284 <b>pass</b><br>power_mean 0.0217 / 0.5 <b>pass</b><br>power_peak 0.5502 / 1.5 <b>pass</b>                                                             |
| detumble_ais_bangbang | 3U AIS — detumble, B-dot law bangbang                       | TRN-P-3U-AIS     | 1.722e+04    | detumble_time 54.02 / 284 <b>pass</b><br>power_mean 0.4366 / 0.5 <b>pass</b><br>power_peak 0.717 / 1.5 <b>pass</b>                                                              |
| detumble_ais_mag      | 3U AIS — detumble, B-dot law mag                            | TRN-P-3U-AIS     | 1.722e+04    | detumble_time 41.16 / 284 <b>pass</b><br>power_mean 0.07839 / 0.5 <b>pass</b><br>power_peak 0.6377 / 1.5 <b>pass</b>                                                            |
| detumble_img          | 3U imaging — B-dot detumble from a 10 deg/s tumble          | TRN-P-3U-IMG     | 1.722e+04    | detumble_time 47.54 / 284 <b>pass</b>                                                                                                                                           |
| fault_coil_ais        | 3U AIS — X coil fails during detumble                       | TRN-P-3U-AIS     | 1.722e+04    | detumble_time 82.17 / 284 <b>pass</b><br>power_mean 0.01821 / 0.5 <b>pass</b><br>power_peak 0.5502 / 1.5 <b>pass</b>                                                            |
| fault_gimbal_cmgy     | 3U imaging — one CMG gimbal sticks during fine hold         | TRN-P-3U-CMG     | 5740         | ape_los_p9973 0.005509 / 0.01 <b>pass</b><br>ake_los_p9973 0.002624 / 0.005 <b>pass</b><br>rate_stability_p9973 0.002963 / 0.001 <b>FAIL</b><br>power_mean 1.61 / 2 <b>pass</b> |
| fault_gyro_ais        | 3U AIS — gyro bias step of 0.1 deg/s during nadir hold      | TRN-P-3U-AIS     | 1.722e+04    | ape_los_last_orbit 52.78 / 10 <b>FAIL</b><br>ake_los_last_orbit 41.92 / 5 <b>FAIL</b><br>power_mean 0.01056 / 0.5 <b>pass</b>                                                   |

| scenario          | what                                                                     | product          | duration [s] | requirement metrics: value / req                                                                                                                                                   |
|-------------------|--------------------------------------------------------------------------|------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fault_st_img      | 3U imaging — star-tracker head 2 fails during fine hold                  | TRN-P-3U-IMG     | 5740         | ape_los_p9973 0.1022 / 0.01 <b>FAIL</b><br>ake_los_p9973 0.1093 / 0.005 <b>FAIL</b><br>rate_stability_p9973 0.003771 / 0.001 <b>FAIL</b><br>power_mean 1.557 / 2 <b>pass</b>       |
| fault_wheel_img   | 3U imaging — reaction wheel 1 fails during fine hold (FDIR + coils)      | TRN-P-3U-IMG     | 5740         | ape_los_p9973 28.62 / 0.01 <b>FAIL</b><br>ake_los_p9973 0.002858 / 0.005 <b>pass</b><br>rate_stability_p9973 0.6244 / 0.001 <b>FAIL</b><br>power_mean 1.213 / 2 <b>pass</b>        |
| fine_hold_cmj     | 3U imaging — fine nadir hold, coils + 4 SGCMG pyramid                    | TRN-P-3U-CMG     | 5740         | ape_los_p9973 0.005021 / 0.01 <b>pass</b><br>ake_los_p9973 0.002626 / 0.005 <b>pass</b><br>rate_stability_p9973 0.002437 / 0.001 <b>FAIL</b><br>power_mean 1.61 / 2 <b>pass</b>    |
| fine_hold_fmr     | 3U imaging — fine nadir hold, coils + fluid momentum rings (IDMAS split) | TRN-P-3U-FMR     | 5740         | ape_los_p9973 0.005689 / 0.01 <b>pass</b><br>ake_los_p9973 0.002631 / 0.005 <b>pass</b><br>rate_stability_p9973 0.004729 / 0.001 <b>FAIL</b><br>power_mean 0.08406 / 2 <b>pass</b> |
| fine_hold_fmr_rcs | 3U imaging — fine nadir hold, coils + fluid rings + cold-gas RCS         | TRN-P-3U-FMR-RCS | 5740         | ape_los_p9973 0.005689 / 0.01 <b>pass</b><br>ake_los_p9973 0.002631 / 0.005 <b>pass</b><br>rate_stability_p9973 0.004729 / 0.001 <b>FAIL</b><br>power_mean 0.08406 / 2 <b>pass</b> |
| fine_hold_img     | 3U imaging — fine nadir hold, reaction wheels + star tracker (0.01 deg)  | TRN-P-3U-IMG     | 5740         | ape_los_p9973 0.01049 / 0.01 <b>FAIL</b><br>ake_los_p9973 0.002608 / 0.005 <b>pass</b><br>rate_stability_p9973 0.003552 / 0.001 <b>FAIL</b><br>power_mean 1.557 / 2 <b>pass</b>    |
| fine_hold_img_lqr | 3U imaging — fine nadir hold, LQR controller                             | TRN-P-3U-IMG     | 5740         | ape_los_p9973 0.004156 / 0.01 <b>pass</b><br>ake_los_p9973 0.002652 / 0.005 <b>pass</b><br>rate_stability_p9973 0.006704 / 0.001 <b>FAIL</b><br>power_mean 1.558 / 2 <b>pass</b>   |
| fine_hold_img_smc | 3U imaging — fine nadir hold, SMC controller                             | TRN-P-3U-IMG     | 5740         | ape_los_p9973 0.01945 / 0.01 <b>FAIL</b><br>ake_los_p9973 0.002649 / 0.005 <b>pass</b><br>rate_stability_p9973 0.003482 / 0.001 <b>FAIL</b><br>power_mean 1.557 / 2 <b>pass</b>    |

| scenario                        | what                                                                                 | product          | duration [s] | requirement metrics: value / req                                                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------------------|------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fine_hold_rw_rcs</code>   | 3U imaging — fine nadir hold, coils + reaction wheels + cold-gas RCS                 | TRN-P-3U-RW-RCS  | 5740         | ape_los_p9973 0.03388 / 0.01 <b>FAIL</b><br>ake_los_p9973 0.002602 / 0.005 <b>pass</b><br>rate_stability_p9973 0.005641 / 0.001 <b>FAIL</b><br>power_mean 1.53 / 2 <b>pass</b>  |
| <code>fine_hold_vscmg</code>    | 3U imaging — fine nadir hold, coils + 4 VSCMG pyramid                                | TRN-P-3U-VSCMG   | 5740         | ape_los_p9973 0.00828 / 0.01 <b>pass</b><br>ake_los_p9973 0.002636 / 0.005 <b>pass</b><br>rate_stability_p9973 0.004762 / 0.001 <b>FAIL</b><br>power_mean 2.014 / 2 <b>FAIL</b> |
| <code>mission_ais</code>        | 3U AIS — integrated chain: detumble then nadir hold (auto mode switch)               | TRN-P-3U-AIS     | 2.87e+04     | detumble_time 37.79 / 284 <b>pass</b><br>ape_los_last_orbit 8.935 / 10 <b>pass</b><br>ake_los_last_orbit 1.188 / 5 <b>pass</b>                                                  |
| <code>mission_cmg</code>        | 3U imaging — tumble, detumble, fine hold: coils + 4 SGCMG pyramid                    | TRN-P-3U-CMG     | 1.722e+04    | detumble_time 57.01 / 284 <b>pass</b><br>ape_los_last_half_orbit_p9973 0.005241 / 0.01 <b>pass</b>                                                                              |
| <code>mission_fmr</code>        | 3U imaging — tumble, detumble, fine hold: coils + fluid momentum rings (IDMAS split) | TRN-P-3U-FMR     | 1.722e+04    | detumble_time 61.33 / 284 <b>pass</b><br>ape_los_last_half_orbit_p9973 0.005954 / 0.01 <b>pass</b>                                                                              |
| <code>mission_fmr_rcs</code>    | 3U imaging — tumble, detumble, fine hold: coils + fluid rings + cold-gas RCS         | TRN-P-3U-FMR-RCS | 1.722e+04    | detumble_time 61.33 / 284 <b>pass</b><br>ape_los_last_half_orbit_p9973 0.005954 / 0.01 <b>pass</b>                                                                              |
| <code>mission_img</code>        | 3U imaging — integrated chain: detumble then fine nadir hold                         | TRN-P-3U-IMG     | 1.722e+04    | detumble_time 51.01 / 284 <b>pass</b><br>ape_los_last_half_orbit_p9973 0.01066 / 0.01 <b>FAIL</b>                                                                               |
| <code>mission_rw_rcs</code>     | 3U imaging — tumble, detumble, fine hold: coils + reaction wheels + cold-gas RCS     | TRN-P-3U-RW-RCS  | 1.722e+04    | detumble_time 51.01 / 284 <b>pass</b><br>ape_los_last_half_orbit_p9973 0.007564 / 0.01 <b>pass</b>                                                                              |
| <code>mission_vscmg</code>      | 3U imaging — tumble, detumble, fine hold: coils + 4 VSCMG pyramid                    | TRN-P-3U-VSCMG   | 1.722e+04    | detumble_time 57.02 / 284 <b>pass</b><br>ape_los_last_half_orbit_p9973 0.009595 / 0.01 <b>pass</b>                                                                              |
| <code>nadir_hold_ais</code>     | 3U AIS — nadir hold with magnetorquers only (10 deg requirement)                     | TRN-P-3U-AIS     | 1.722e+04    | ape_los_last_orbit 12.64 / 10 <b>FAIL</b><br>ake_los_last_orbit 5.491 / 5 <b>FAIL</b><br>power_mean 0.01054 / 0.5 <b>pass</b>                                                   |
| <code>nadir_hold_ais_css</code> | 3U AIS — nadir hold with coarse cosine sun sensors (albedo)                          | TRN-P-3U-AIS-CSS | 1.722e+04    | ape_los_last_orbit 39.44 / 10 <b>FAIL</b><br>ake_los_last_orbit 39.22 / 5 <b>FAIL</b><br>power_mean 0.01064 / 0.5 <b>pass</b>                                                   |
| <code>slew_cmg</code>           | 3U imaging — 30 deg target slew in 60 s, coils + 4 SGCMG pyramid                     | TRN-P-3U-CMG     | 600          | settle_time_after_slew 12.4 / 20 <b>pass</b><br>ape_los_on_target_p9973 0.004233 / 0.01 <b>pass</b>                                                                             |
| <code>slew_fmr</code>           | 3U imaging — 30 deg target slew in 60 s, coils + fluid momentum rings (IDMAS split)  | TRN-P-3U-FMR     | 600          | settle_time_after_slew 12.9 / 20 <b>pass</b><br>ape_los_on_target_p9973 0.006529 / 0.01 <b>pass</b>                                                                             |

| scenario     | what                                                                                  | product                  | duration [s] | requirement metrics: value / req                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------|--------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slew_fmr_rcs | 3U imaging — 30 deg target<br>slew in 60 s, coils + fluid<br>rings + cold-gas RCS     | TRN-P-<br>3U-FMR-<br>RCS | 600          | settle_time_after_slew 12.9 / 20<br><b>pass</b><br>ape_los_on_target_p9973<br>0.006529 / 0.01 <b>pass</b>                                                                                                                                    |
| slew_img     | 3U imaging — 30 deg target<br>slew in 60 s and settle                                 | TRN-P-<br>3U-IMG         | 600          | settle_time_after_slew 12.1 / 20<br><b>pass</b><br>ape_los_on_target_p9973<br>0.007925 / 0.01 <b>pass</b>                                                                                                                                    |
| slew_img_lqr | 3U imaging — 30 deg slew,<br>LQR controller                                           | TRN-P-<br>3U-IMG         | 600          | settle_time_after_slew 0 / 20 <b>pass</b><br>ape_los_on_target_p9973<br>0.004439 / 0.01 <b>pass</b>                                                                                                                                          |
| slew_img_smc | 3U imaging — 30 deg slew,<br>SMC controller                                           | TRN-P-<br>3U-IMG         | 600          | settle_time_after_slew — / 20<br><b>FAIL</b><br>ape_los_on_target_p9973 0.0178 /<br>0.01 <b>FAIL</b>                                                                                                                                         |
| slew_rw_rcs  | 3U imaging — 30 deg target<br>slew in 60 s, coils + reaction<br>wheels + cold-gas RCS | TRN-P-<br>3U-RW-<br>RCS  | 600          | settle_time_after_slew 12.1 / 20<br><b>pass</b><br>ape_los_on_target_p9973<br>0.007922 / 0.01 <b>pass</b>                                                                                                                                    |
| slew_vscmg   | 3U imaging — 30 deg target<br>slew in 60 s, coils + 4<br>VSCMG pyramid                | TRN-P-<br>3U-<br>VSCMG   | 600          | settle_time_after_slew 13.2 / 20<br><b>pass</b><br>ape_los_on_target_p9973 0.01083<br>/ 0.01 <b>FAIL</b>                                                                                                                                     |
| sun_spin_ais | 3U AIS — coils-only Sun<br>acquisition: detumble, spin-<br>up, Sun-pointing spin      | TRN-P-<br>3U-AIS         | 1.722e+04    | sun_spin_entry 142.4 / 95 <b>FAIL</b><br>sun_angle_last_orbit 93.84 / 20<br><b>FAIL</b><br>spin_rate_error_last_orbit 1.116 /<br>0.5 <b>FAIL</b><br>sun_spin_share_last_orbit 13.75 /<br>90 <b>FAIL</b><br>power_mean 0.45 / 0.5 <b>pass</b> |

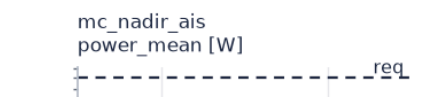
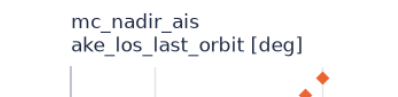
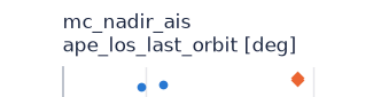
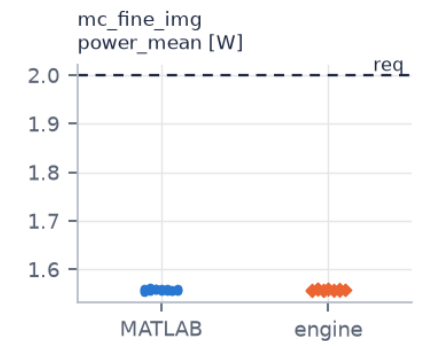
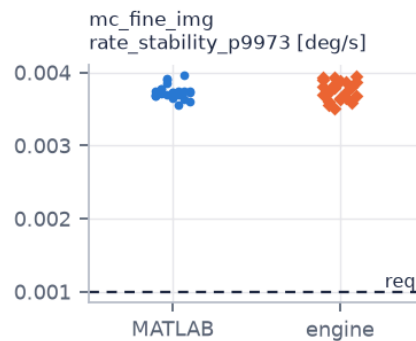
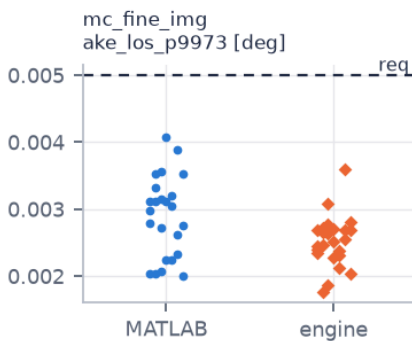
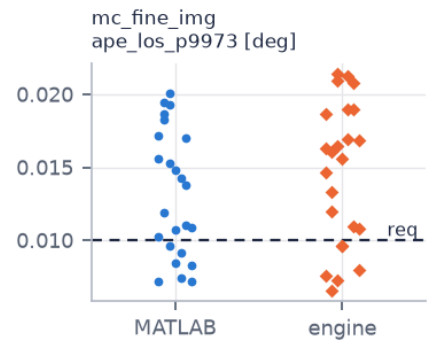
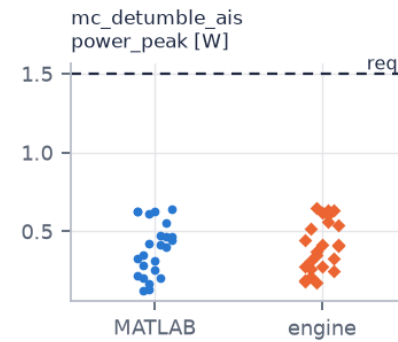
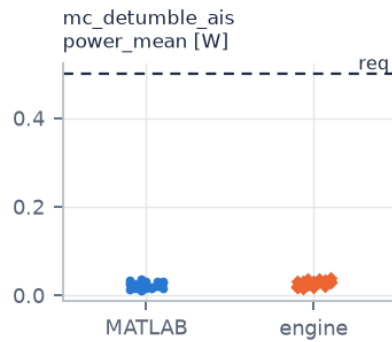
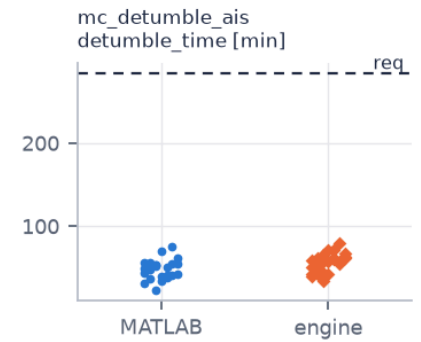
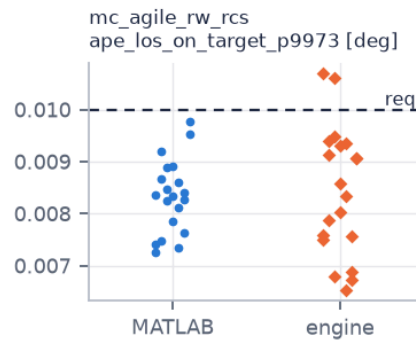
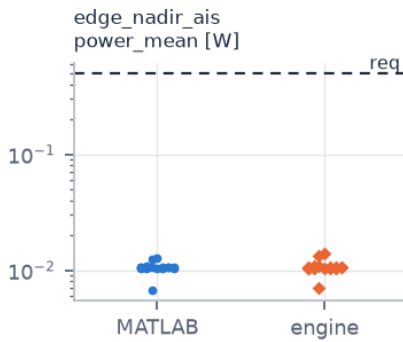
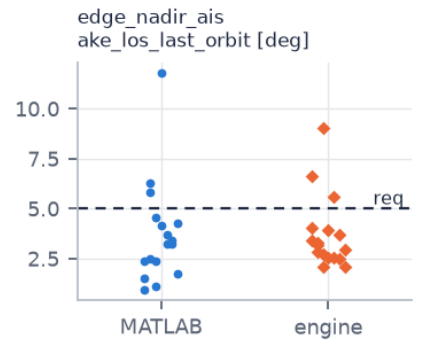
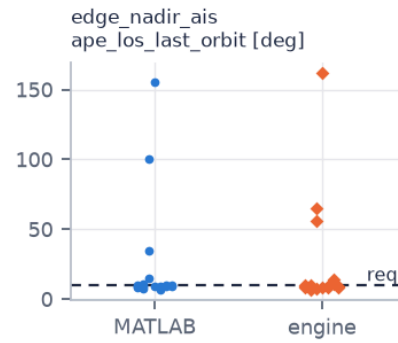
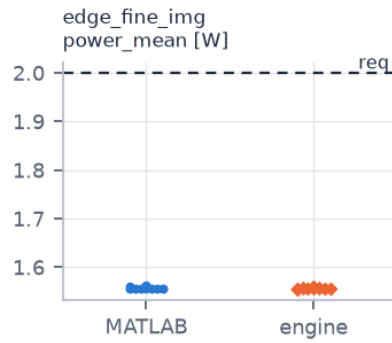
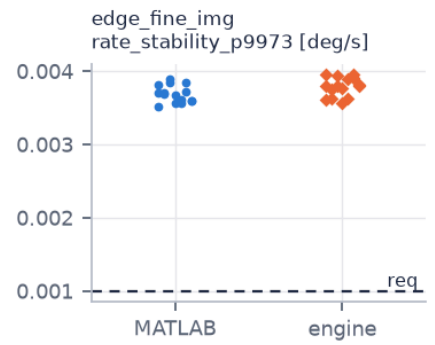
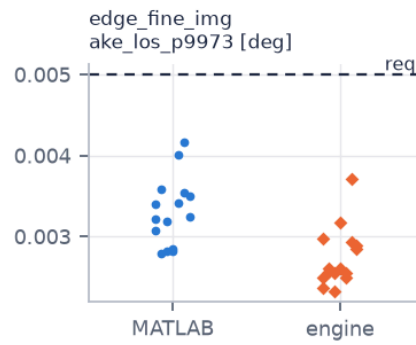
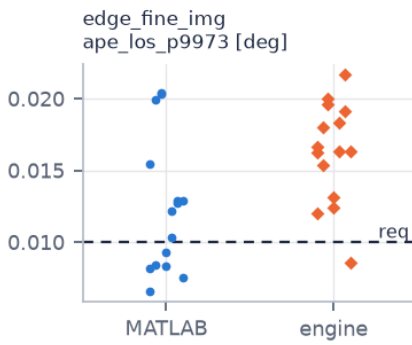
## 6. Monte Carlo and edge campaigns vs the MATLAB twin

Every campaign of `campaigns/*.toml` flown on the Rust engine with the draws of `asils.campaign.draw` (same dispersions, bounds, run count, per-run seeds), next to the MATLAB twin's campaign. Monte Carlo realisations differ (different random streams) so distributions are compared; edge runs are compared run by run. The engine's flight software keeps the nominal inertia while the plant is dispersed; the MATLAB twin's laws read the dispersed inertia.

| campaign        | type       | metric                  | req   | MATLAB<br>mean $\pm$ std   | MATLAB<br>pass | engine<br>mean $\pm$ std   | engine<br>pass |
|-----------------|------------|-------------------------|-------|----------------------------|----------------|----------------------------|----------------|
| edge_fine_img   | edge       | ape_los_p9973           | 0.01  | 0.01236 $\pm$<br>0.00475   | 40 %           | 0.01625 $\pm$<br>0.00351   | 7 %            |
| edge_fine_img   | edge       | ake_los_p9973           | 0.005 | 0.003307 $\pm$<br>0.000418 | 100 %          | 0.002739 $\pm$<br>0.000362 | 100 %          |
| edge_fine_img   | edge       | rate_stability_p9973    | 0.001 | 0.003689 $\pm$<br>0.000116 | 0 %            | 0.003788 $\pm$<br>0.00013  | 0 %            |
| edge_fine_img   | edge       | power_mean              | 2     | 1.558 $\pm$<br>0.00216     | 100 %          | 1.557 $\pm$<br>0.00112     | 100 %          |
| edge_nadir_ais  | edge       | ape_los_last_orbit      | 10    | 24.91 $\pm$ 40.4           | 59 %           | 24.19 $\pm$ 39.4           | 65 %           |
| edge_nadir_ais  | edge       | ake_los_last_orbit      | 5     | 3.712 $\pm$ 2.58           | 82 %           | 3.713 $\pm$ 1.82           | 82 %           |
| edge_nadir_ais  | edge       | power_mean              | 0.5   | 0.01057 $\pm$<br>0.00121   | 100 %          | 0.01069 $\pm$<br>0.00144   | 100 %          |
| mc_agile_rw_rcs | montecarlo | ape_los_on_target_p9973 | 0.01  | 0.008337 $\pm$<br>0.000712 | 100 %          | 0.008427 $\pm$<br>0.00124  | 90 %           |
| mc_detumble_ais | montecarlo | detumble_time           | 284   | 48.5 $\pm$ 12              | 100 %          | 56.29 $\pm$ 11.4           | 100 %          |
| mc_detumble_ais | montecarlo | power_mean              | 0.5   | 0.02267 $\pm$<br>0.00701   | 100 %          | 0.02764 $\pm$<br>0.00684   | 100 %          |
| mc_detumble_ais | montecarlo | power_peak              | 1.5   | 0.3902 $\pm$<br>0.169      | 100 %          | 0.402 $\pm$ 0.16           | 100 %          |
| mc_fine_img     | montecarlo | ape_los_p9973           | 0.01  | 0.01315 $\pm$<br>0.00434   | 29 %           | 0.01456 $\pm$<br>0.00486   | 25 %           |
| mc_fine_img     | montecarlo | ake_los_p9973           | 0.005 | 0.002894 $\pm$<br>0.000605 | 100 %          | 0.002523 $\pm$<br>0.000383 | 100 %          |
| mc_fine_img     | montecarlo | rate_stability_p9973    | 0.001 | 0.003738 $\pm$<br>9.57e-05 | 0 %            | 0.003755 $\pm$<br>0.00013  | 0 %            |
| mc_fine_img     | montecarlo | power_mean              | 2     | 1.558 $\pm$<br>0.0021      | 100 %          | 1.558 $\pm$<br>0.00149     | 100 %          |
| mc_nadir_ais    | montecarlo | ape_los_last_orbit      | 10    | 89.69 $\pm$ 38.5           | 0 %            | 92.98 $\pm$ 42.3           | 0 %            |
| mc_nadir_ais    | montecarlo | ake_los_last_orbit      | 5     | 3.3 $\pm$ 1.73             | 75 %           | 2.829 $\pm$ 1.69           | 88 %           |
| mc_nadir_ais    | montecarlo | power_mean              | 0.5   | 0.01145 $\pm$<br>0.0025    | 100 %          | 0.01004 $\pm$<br>0.0029    | 100 %          |
| mc_slew_cmg     | montecarlo | settle_time_after_slew  | 20    | 13.14 $\pm$ 0.46           | 100 %          | 13.03 $\pm$<br>0.785       | 100 %          |
| mc_slew_cmg     | montecarlo | ape_los_on_target_p9973 | 0.01  | 0.003931 $\pm$<br>0.000635 | 100 %          | 0.003761 $\pm$<br>0.000349 | 100 %          |

| campaign    | type       | metric                  | req  | MATLAB<br>mean ± std | MATLAB<br>pass | engine<br>mean ± std | engine<br>pass |
|-------------|------------|-------------------------|------|----------------------|----------------|----------------------|----------------|
| mc_slew_img | montecarlo | settle_time_after_slew  | 20   | 13.03 ±<br>0.959     | 100 %          | 13.84 ± 4.76         | 98 %           |
| mc_slew_img | montecarlo | ape_los_on_target_p9973 | 0.01 | 0.00815 ±<br>0.00115 | 92 %           | 0.007886 ±<br>0.001  | 95 %           |

• MATLAB twin    ♦ Rust engine



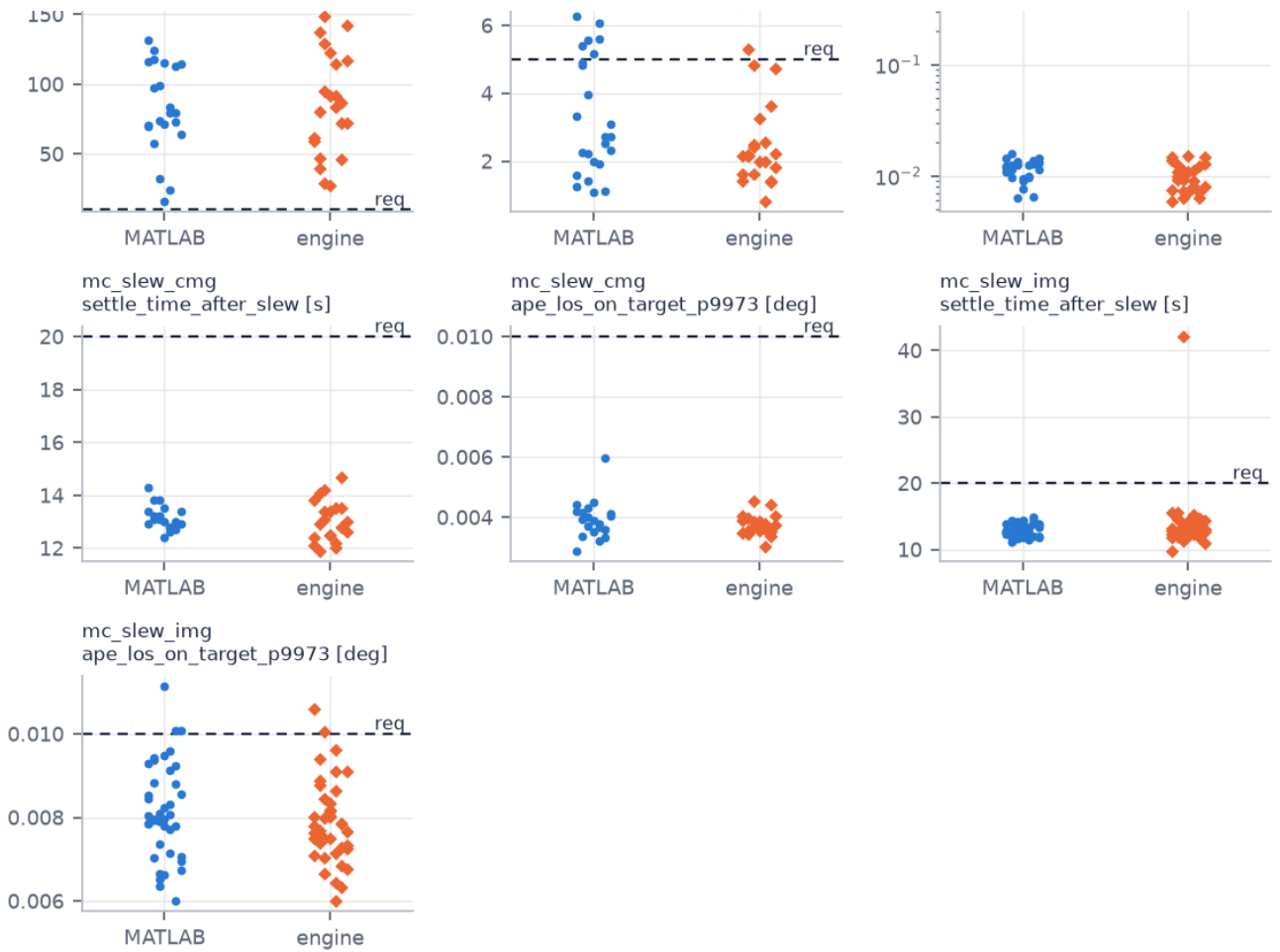


Figure 3. Every run of every campaign, requirement metrics: MATLAB twin (circles) and Rust engine (diamonds), dashed line = requirement.

## 7. Engine vs MATLAB twin parity

Single runs, engine vs MATLAB twin, same scenario, case and product: **95 of 107** requirement verdicts agree. The truth environment (orbit, Sun, Moon, shadow, density, field) of the engine's POP port is identical to the twin's on every filed sample of every scenario; what differs is the random stream (unit errors, initial tumble), so single-run metrics scatter inside the distributions of section 6. The differences traced to their cause are listed in results/ENGINE\_PARITY.md.

| scenario              | samples | max  r  difference [m] | max density ratio – 1 | max Sun-direction difference | max  B |
|-----------------------|---------|------------------------|-----------------------|------------------------------|--------|
| agile_slew_cmgy       | 6001    | 0                      | 0                     | 0                            |        |
| agile_slew_fmr_rcs    | 6001    | 0                      | 0                     | 0                            |        |
| agile_slew_img        | 6001    | 0                      | 0                     | 0                            |        |
| agile_slew_rw_rcs     | 6001    | 0                      | 0                     | 0                            |        |
| agile_slew_vscmg      | 6001    | 0                      | 0                     | 0                            |        |
| detumble_ais          | 17221   | 0                      | 0                     | 0                            |        |
| detumble_ais_bangbang | 17221   | 0                      | 0                     | 0                            |        |
| detumble_ais_mag      | 17221   | 0                      | 0                     | 0                            |        |
| detumble_img          | 17221   | 0                      | 0                     | 0                            |        |
| fault_coil_ais        | 17221   | 0                      | 0                     | 0                            |        |
| fault_gimbal_cmgy     | 57401   | 0                      | 0                     | 0                            |        |
| fault_gyro_ais        | 17221   | 0                      | 0                     | 0                            |        |
| fault_st_img          | 57401   | 0                      | 0                     | 0                            |        |
| fault_wheel_img       | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_cmgy        | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_fmr         | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_fmr_rcs     | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_img         | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_img_lqr     | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_img_smc     | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_rw_rcs      | 57401   | 0                      | 0                     | 0                            |        |
| fine_hold_vscmg       | 57401   | 0                      | 0                     | 0                            |        |
| mission_ais           | 28701   | 0                      | 0                     | 0                            |        |
| mission_cmgy          | 17221   | 0                      | 0                     | 0                            |        |
| mission_fmr           | 17221   | 0                      | 0                     | 0                            |        |
| mission_fmr_rcs       | 17221   | 0                      | 0                     | 0                            |        |
| mission_img           | 17221   | 0                      | 0                     | 0                            |        |
| mission_rw_rcs        | 17221   | 0                      | 0                     | 0                            |        |
| mission_vscmg         | 17221   | 0                      | 0                     | 0                            |        |
| nadir_hold_ais        | 17221   | 0                      | 0                     | 0                            |        |

| scenario              | samples | max  r  difference [m] | max density ratio – 1 | max Sun-direction difference | max  B |
|-----------------------|---------|------------------------|-----------------------|------------------------------|--------|
| nadir_hold_ais_css    | 17221   | 0                      | 0                     | 0                            |        |
| slew_cmgy             | 6001    | 0                      | 0                     | 0                            |        |
| slew_fmry             | 6001    | 0                      | 0                     | 0                            |        |
| slew_fmry_rcs         | 6001    | 0                      | 0                     | 0                            |        |
| slew_img              | 6001    | 0                      | 0                     | 0                            |        |
| slew_img_lqr          | 6001    | 0                      | 0                     | 0                            |        |
| slew_img_smc          | 6001    | 0                      | 0                     | 0                            |        |
| slew_rw_rcs           | 6001    | 0                      | 0                     | 0                            |        |
| slew_vscmg            | 6001    | 0                      | 0                     | 0                            |        |
| sun_spin_ais          | 17221   | 0                      | 0                     | 0                            |        |
| agile_slew_cmgy       | 55      | 2.17                   | 26x                   |                              |        |
| agile_slew_fmry_rcs   | 51      | 2.20                   | 23x                   |                              |        |
| agile_slew_img        | 48      | 1.53                   | 31x                   |                              |        |
| agile_slew_rw_rcs     | 55      | 1.52                   | 36x                   |                              |        |
| agile_slew_vscmg      | 61      | 0.20                   | 300x                  |                              |        |
| detumble_ais          | 496     | 0.97                   | 511x                  |                              |        |
| detumble_ais_bangbang | 848     | 0.74                   | 1152x                 |                              |        |
| detumble_ais_mag      | 526     | 1.34                   | 392x                  |                              |        |
| detumble_img          | 972     | 4.09                   | 237x                  |                              |        |
| fault_coil_ais        | 533     | 0.59                   | 908x                  |                              |        |
| fault_gimbal_cmgy     | 465     | 2.02                   | 230x                  |                              |        |
| fault_gyro_ais        | 579     | 1.20                   | 481x                  |                              |        |
| fault_st_img          | 389     | 1.74                   | 223x                  |                              |        |
| fault_wheel_img       | 414     | 2.15                   | 192x                  |                              |        |
| fine_hold_cmgy        | 468     | 2.58                   | 182x                  |                              |        |
| fine_hold_fmry        | 659     | 1.77                   | 371x                  |                              |        |
| fine_hold_fmry_rcs    | 523     | 2.28                   | 230x                  |                              |        |
| fine_hold_img         | 390     | 1.38                   | 282x                  |                              |        |
| fine_hold_img_lqr     | 388     | 1.91                   | 203x                  |                              |        |
| fine_hold_img_smc     | 388     | 1.37                   | 284x                  |                              |        |
| fine_hold_rw_rcs      | 394     | 1.72                   | 229x                  |                              |        |
| fine_hold_vscmg       | 476     | 2.23                   | 213x                  |                              |        |
| mission_ais           | 906     | 0.78                   | 1158x                 |                              |        |
| mission_cmgy          | 1421    | 5.67                   | 251x                  |                              |        |
| mission_fmry          | 1622    | 4.49                   | 361x                  |                              |        |
| mission_fmry_rcs      | 1619    | 4.82                   | 336x                  |                              |        |
| mission_img           | 1235    | 4.01                   | 308x                  |                              |        |
| mission_rw_rcs        | 1263    | 4.95                   | 255x                  |                              |        |

| scenario           | samples | max  r  difference [m] | max density ratio – 1 | max Sun-direction difference | max  B |
|--------------------|---------|------------------------|-----------------------|------------------------------|--------|
| mission_vscmg      | 1461    | 5.92                   | 247x                  |                              |        |
| nadir_hold_ais     | 569     | 0.64                   | 892x                  |                              |        |
| nadir_hold_ais_css | 550     | 0.66                   | 836x                  |                              |        |
| slew_cmg           | 54      | 0.54                   | 100x                  |                              |        |
| slew_fmr           | 79      | 0.22                   | 358x                  |                              |        |
| slew_fmr_rcs       | 56      | 0.22                   | 249x                  |                              |        |
| slew_img           | 46      | 0.62                   | 74x                   |                              |        |
| slew_img_lqr       | 46      | 0.59                   | 77x                   |                              |        |
| slew_img_smc       | 45      | 0.24                   | 192x                  |                              |        |
| slew_rw_rcs        | 46      | 0.18                   | 260x                  |                              |        |
| slew_vscmg         | 56      | 0.33                   | 168x                  |                              |        |
| sun_spin_ais       | 501     | 0.62                   | 809x                  |                              |        |

Verdicts that differ (all traced; see ENGINE\_PARITY.md):

| scenario          | metric                        | MATLAB   | engine   | req  | MATLAB | engine |
|-------------------|-------------------------------|----------|----------|------|--------|--------|
| fine_hold_fmr     | power_mean                    | 4.348    | 0.08406  | 2    | FAIL   | pass   |
| fine_hold_fmr_rcs | power_mean                    | 4.367    | 0.08406  | 2    | FAIL   | pass   |
| fine_hold_img     | ape_los_p9973                 | 0.009166 | 0.01049  | 0.01 | pass   | FAIL   |
| fine_hold_rw_rcs  | ape_los_p9973                 | 0.008375 | 0.03388  | 0.01 | pass   | FAIL   |
| mission_img       | ape_los_last_half_orbit_p9973 | 0.007627 | 0.01066  | 0.01 | pass   | FAIL   |
| mission_vscmg     | ape_los_last_half_orbit_p9973 | 0.01111  | 0.009595 | 0.01 | FAIL   | pass   |
| nadir_hold_ais    | ape_los_last_orbit            | 7.248    | 12.64    | 10   | pass   | FAIL   |
| nadir_hold_ais    | ake_los_last_orbit            | 3.864    | 5.491    | 5    | pass   | FAIL   |
| slew_img_smc      | settle_time_after_slew        | 3.2      | —        | 20   | pass   | FAIL   |
| sun_spin_ais      | sun_spin_entry                | 40.44    | 142.4    | 95   | pass   | FAIL   |
| sun_spin_ais      | sun_angle_last_orbit          | 6.503    | 93.84    | 20   | pass   | FAIL   |
| sun_spin_ais      | spin_rate_error_last_orbit    | 0.04568  | 1.116    | 0.5  | pass   | FAIL   |

## 8. Soft OILS

Every scenario flown twice on the engine: SILS (C flight software in-process) and soft OILS (the flight software compiled for the OBC, arm-none-eabi-gcc -mcpu=cortex-m4 -mfloat-abi=hard, running as firmware on QEMU mps2-an386 behind adcs-link/1). In soft OILS each command reaches the actuators only after the OBC's sensor reads, its execution and its CAN frames; the execution is the step's exact instruction count (QEMU plugin)  $\times$  CPI 1.25 / 168 MHz. **105 of 107** requirement verdicts agree between SILS and soft OILS over 40 scenarios; runs are deterministic.

| scenario              | period [ms] | instructions/step mean / max | exec max [ms] | latency mean / max [ms] | C |
|-----------------------|-------------|------------------------------|---------------|-------------------------|---|
| agile_slew_cmgy       | 100         | 3.5e+05 / 8.94e+05           | 6.65          | 4.2 / 8.24              |   |
| agile_slew_fmr_rcs    | 100         | 3.01e+05 / 8.45e+05          | 6.29          | 3.31 / 7.36             |   |
| agile_slew_img        | 100         | 2.94e+05 / 8.44e+05          | 6.28          | 3.14 / 7.22             |   |
| agile_slew_rw_rcs     | 100         | 3.04e+05 / 8.51e+05          | 6.33          | 3.34 / 7.4              |   |
| agile_slew_vscmg      | 100         | 3.68e+05 / 9.17e+05          | 6.82          | 4.33 / 8.41             |   |
| detumble_ais          | 200         | 1.19e+05 / 4.64e+05          | 3.45          | 1.44 / 4                |   |
| detumble_ais_bangbang | 200         | 1.18e+05 / 4.64e+05          | 3.45          | 1.43 / 4                |   |
| detumble_ais_mag      | 200         | 1.19e+05 / 4.64e+05          | 3.45          | 1.44 / 4                |   |
| detumble_img          | 100         | 8.12e+04 / 4.89e+05          | 3.64          | 1.55 / 4.58             |   |
| fault_coil_ais        | 200         | 1.19e+05 / 4.64e+05          | 3.45          | 1.44 / 4                |   |
| fault_gimbal_cmgy     | 100         | 3.62e+05 / 8.98e+05          | 6.68          | 4.29 / 8.27             |   |
| fault_gyro_ais        | 200         | 1.93e+05 / 6.37e+05          | 4.74          | 1.99 / 5.3              |   |
| fault_st_img          | 100         | 2.54e+05 / 8.42e+05          | 6.26          | 2.84 / 7.21             |   |
| fault_wheel_img       | 100         | 2.9e+05 / 8.44e+05           | 6.28          | 3.1 / 7.22              |   |
| fine_hold_cmgy        | 100         | 3.62e+05 / 8.97e+05          | 6.68          | 4.29 / 8.27             |   |
| fine_hold_fmr         | 100         | 3.1e+05 / 8.45e+05           | 6.29          | 3.25 / 7.23             |   |
| fine_hold_fmr_rcs     | 100         | 3.13e+05 / 8.48e+05          | 6.31          | 3.4 / 7.39              |   |
| fine_hold_img         | 100         | 3.1e+05 / 8.45e+05           | 6.29          | 3.25 / 7.23             |   |
| fine_hold_img_lqr     | 100         | 3.1e+05 / 8.46e+05           | 6.29          | 3.25 / 7.24             |   |
| fine_hold_img_smc     | 100         | 3.13e+05 / 8.48e+05          | 6.31          | 3.27 / 7.26             |   |
| fine_hold_rw_rcs      | 100         | 3.15e+05 / 8.52e+05          | 6.34          | 3.42 / 7.42             |   |
| fine_hold_vscmg       | 100         | 3.83e+05 / 9.19e+05          | 6.84          | 4.44 / 8.43             |   |
| mission_ais           | 200         | 1.86e+05 / 7.58e+05          | 5.64          | 1.94 / 6.2              |   |
| mission_cmgy          | 100         | 3.05e+05 / 8.97e+05          | 6.67          | 3.87 / 8.27             |   |
| mission_fmr           | 100         | 2.61e+05 / 8.46e+05          | 6.3           | 2.89 / 7.24             |   |
| mission_fmr_rcs       | 100         | 2.64e+05 / 8.49e+05          | 6.32          | 3.04 / 7.39             |   |
| mission_img           | 100         | 2.68e+05 / 8.45e+05          | 6.29          | 2.94 / 7.23             |   |
| mission_rw_rcs        | 100         | 2.74e+05 / 8.55e+05          | 6.36          | 3.12 / 7.43             |   |
| mission_vscmg         | 100         | 3.23e+05 / 9.18e+05          | 6.83          | 3.99 / 8.42             |   |

| scenario           | period [ms] | instructions/step mean / max | exec max [ms] | latency mean / max [ms] | C |
|--------------------|-------------|------------------------------|---------------|-------------------------|---|
| nadir_hold_ais     | 200         | 1.93e+05 / 6.37e+05          | 4.74          | 1.99 / 5.3              |   |
| nadir_hold_ais_css | 200         | 1.93e+05 / 6.38e+05          | 4.74          | 1.99 / 5.3              |   |
| slew_cmng          | 100         | 3.69e+05 / 8.94e+05          | 6.65          | 4.34 / 8.24             |   |
| slew_fmr           | 100         | 3.17e+05 / 8.41e+05          | 6.26          | 3.3 / 7.2               |   |
| slew_fmr_rcs       | 100         | 3.19e+05 / 8.44e+05          | 6.28          | 3.45 / 7.36             |   |
| slew_img           | 100         | 3.16e+05 / 8.42e+05          | 6.26          | 3.3 / 7.21              |   |
| slew_img_lqr       | 100         | 3.17e+05 / 8.42e+05          | 6.27          | 3.3 / 7.21              |   |
| slew_img_smc       | 100         | 3.19e+05 / 8.45e+05          | 6.29          | 3.32 / 7.23             |   |
| slew_rw_rcs        | 100         | 3.21e+05 / 8.47e+05          | 6.31          | 3.46 / 7.38             |   |
| slew_vscmg         | 100         | 3.89e+05 / 9.15e+05          | 6.81          | 4.49 / 8.4              |   |
| sun_spin_ais       | 200         | 1.21e+05 / 4.65e+05          | 3.46          | 1.45 / 4.01             |   |

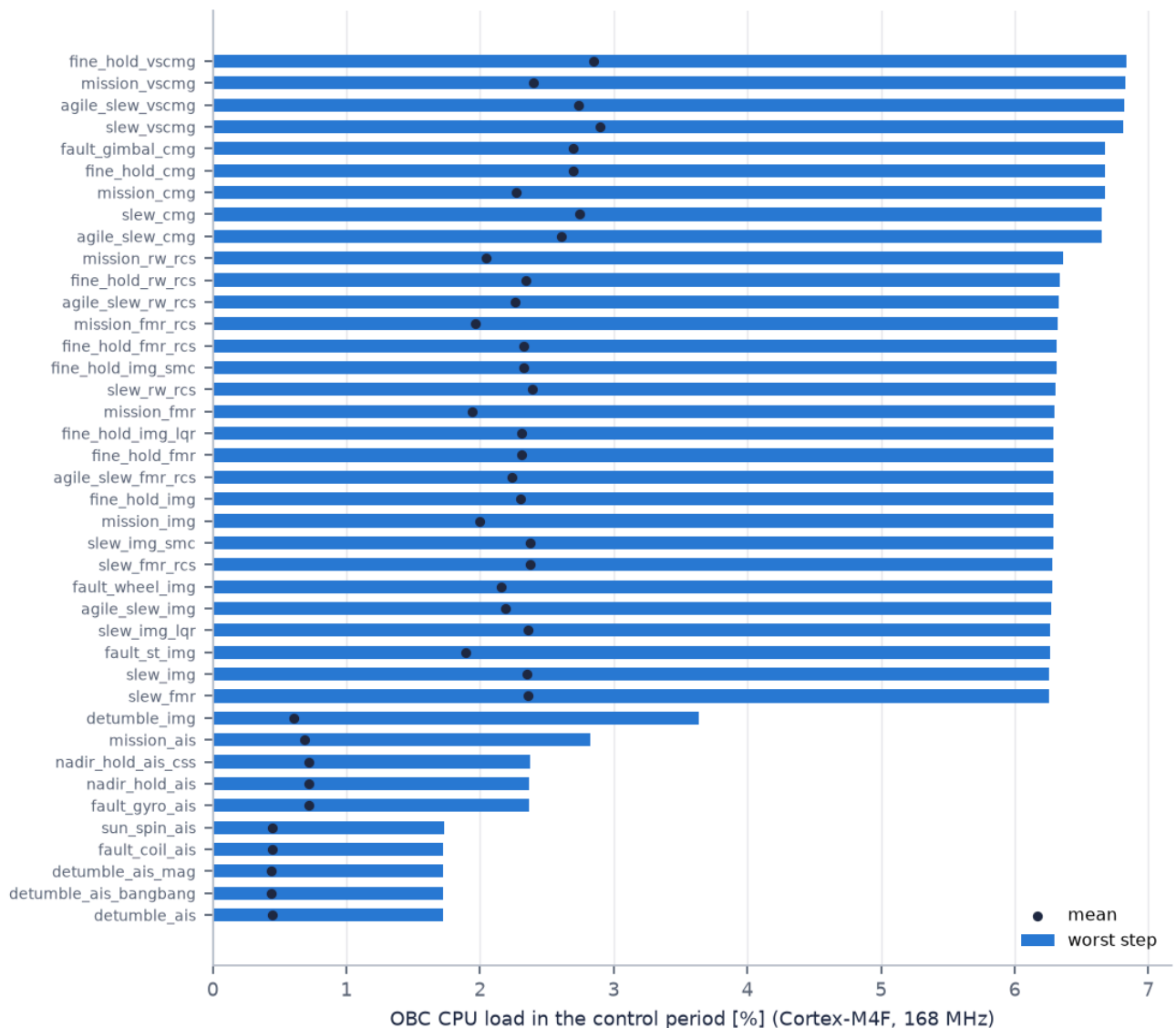


Figure 4. OBC load per scenario: worst step (bar) and mean (dot).

Verdicts that change between SILS and soft OILS (the command latency's effect):

| scenario         | metric                  | req  | SILS    | soft OILS | SILS | soft OILS |
|------------------|-------------------------|------|---------|-----------|------|-----------|
| agile_slew_vscmg | ape_los_on_target_p9973 | 0.01 | 0.01008 | 0.009977  | FAIL | pass      |
| sun_spin_ais     | sun_spin_entry          | 95   | 142.4   | 59.66     | FAIL | pass      |

## 9. Virtual OBC bit-identity

Zero-latency lockstep (no timing model): the flight software as a host process and as QEMU Cortex-M4F firmware, C and Rust, reproduces the in-process trajectory bit for bit: **35 of 35** runs bit-identical.

| scenario          | reference         | virtual OBC  | result        | wall [s] |
|-------------------|-------------------|--------------|---------------|----------|
| detumble_ais      | c (in-process)    | obc-posix    | bit-identical | 0.3      |
| detumble_ais      | rust (in-process) | obc-posix-rs | bit-identical | 0.2      |
| detumble_ais      | c (in-process)    | qemu         | bit-identical | 18.2     |
| detumble_ais      | rust (in-process) | qemu-rs      | bit-identical | 17.4     |
| detumble_ais      | c (in-process)    | rust         | bit-identical | 0.2      |
| mission_ais       | c (in-process)    | obc-posix    | bit-identical | 0.2      |
| mission_ais       | rust (in-process) | obc-posix-rs | bit-identical | 0.2      |
| mission_ais       | c (in-process)    | qemu         | bit-identical | 14.2     |
| mission_ais       | rust (in-process) | qemu-rs      | bit-identical | 15.4     |
| mission_ais       | c (in-process)    | rust         | bit-identical | 0.1      |
| fine_hold_img     | c (in-process)    | obc-posix    | bit-identical | 0.6      |
| fine_hold_img     | rust (in-process) | obc-posix-rs | bit-identical | 0.5      |
| fine_hold_img     | c (in-process)    | qemu         | bit-identical | 55.9     |
| fine_hold_img     | rust (in-process) | qemu-rs      | bit-identical | 61.1     |
| fine_hold_img     | c (in-process)    | rust         | bit-identical | 0.3      |
| slew_img          | c (in-process)    | obc-posix    | bit-identical | 0.6      |
| slew_img          | rust (in-process) | obc-posix-rs | bit-identical | 0.5      |
| slew_img          | c (in-process)    | qemu         | bit-identical | 62.8     |
| slew_img          | rust (in-process) | qemu-rs      | bit-identical | 63.3     |
| slew_img          | c (in-process)    | rust         | bit-identical | 0.3      |
| fine_hold_cmng    | c (in-process)    | obc-posix    | bit-identical | 0.6      |
| fine_hold_cmng    | rust (in-process) | obc-posix-rs | bit-identical | 0.6      |
| fine_hold_cmng    | c (in-process)    | qemu         | bit-identical | 64.2     |
| fine_hold_cmng    | rust (in-process) | qemu-rs      | bit-identical | 68.9     |
| fine_hold_cmng    | c (in-process)    | rust         | bit-identical | 0.3      |
| fine_hold_fmr_rcs | c (in-process)    | obc-posix    | bit-identical | 0.4      |
| fine_hold_fmr_rcs | rust (in-process) | obc-posix-rs | bit-identical | 0.4      |
| fine_hold_fmr_rcs | c (in-process)    | qemu         | bit-identical | 57.5     |
| fine_hold_fmr_rcs | rust (in-process) | qemu-rs      | bit-identical | 57.1     |
| fine_hold_fmr_rcs | c (in-process)    | rust         | bit-identical | 0.5      |

| scenario     | reference         | virtual OBC  | result        | wall [s] |
|--------------|-------------------|--------------|---------------|----------|
| sun_spin_ais | c (in-process)    | obc-posix    | bit-identical | 0.2      |
| sun_spin_ais | rust (in-process) | obc-posix-rs | bit-identical | 0.2      |
| sun_spin_ais | c (in-process)    | qemu         | bit-identical | 15.3     |
| sun_spin_ais | rust (in-process) | qemu-rs      | bit-identical | 13.6     |
| sun_spin_ais | c (in-process)    | rust         | bit-identical | 0.1      |

## 10. Open items

- **ais\_3u**: the selected `mtq_fmr` leaves budget: `mass_kg` 1.6 > 0.35; budget: `volume_L` 0.537 > 0.3. Either the case relaxes these (several are marked UNCONFIRMED in the case file) or the next design lever is needed.
- **ais\_img\_3u**: the selected `mtq_fmr` leaves budget: `mass_kg` 1.75 > 1. Either the case relaxes these (several are marked UNCONFIRMED in the case file) or the next design lever is needed.
- Bang-bang B-dot (fixed): pure sign switching limit-cycled around the 0.5 deg/s exit rate in 4 of 12 seeds with or without OBC latency; a boundary layer (4 x the B-dot gain inside it) detumbles 12 of 12 at 0, 4 and 8 ms. C, Rust, MATLAB and pseudocode updated.
- Rate stability (imaging, 0.001 deg/s, UNCONFIRMED): limited by gyro noise for every actuator (0.0025-0.004 deg/s with TRN-GYRO-P1); a gyro with 0.3 x the noise (fibre-optic class, ~+0.14 kg, +0.7 W) reaches 0.0009 deg/s. The loop proposes it; the mass budget decides.
- Fluid loop fine pointing: limited by the loop's flow sensor, not the pump. At 0.2 mm/s (1 sigma) the loop reaches ~0.004 deg APE, as wheels do; this is a requirement on the in-house flow sensor.
- Sun spin: 17 of 24 seeds pass with and without the OBC; failures are Sun-spin entry with the Sun near the XY plane tripping the 1 deg/s exit guard during the L2 precession transient -- a tuning trade (`fsw.sun_spin_perp_out_dps`, `fsw.sun_spin_dwell_out_s`).
- Soft OILS models the OBC's CPU and buses; real OILS still needs the board target (`fsw/targets/<board>/main.c`: UART/Ethernet, clock, linker script, modelled on `fsw/targets/qemu-mps2`) and `--realtime` on `adcs-link/1`.